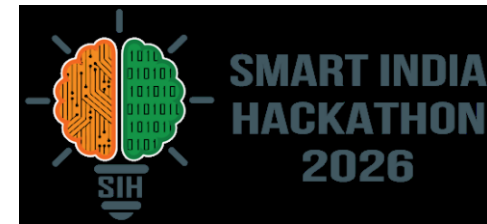


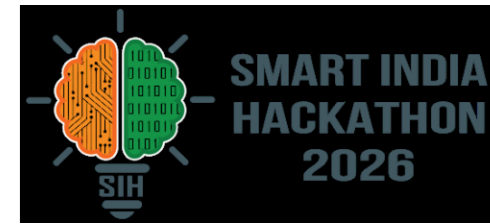
SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – **SIH 26160**
- Problem Statement Title- **AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework**
- Theme- **Blockchain & Cybersecurity**
- PS Category- **Software**
- Team ID- **SIH 163**
- Team Name (Registered on portal)- **Highlanders**

PROPOSED SOLUTION



◆ IPsec Sentinel — passive IPsec assessment with auditable evidence

Real-world issue:

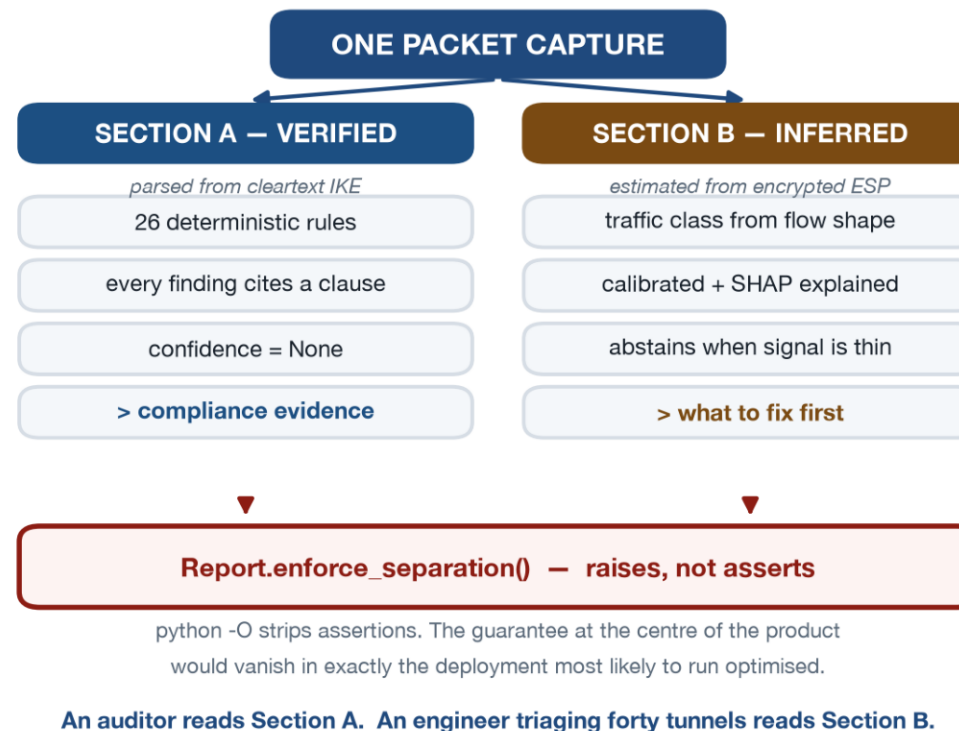
Wireshark decodes an IKE exchange but has no opinion about it. Weak IPsec survives for years because nothing judges what it sees.

How it addresses the problem:

26 deterministic rules, each citing the clause it applies, across NIST, CNSA, BSI, RFC 8221/8247, ITSAR and CERT-In.

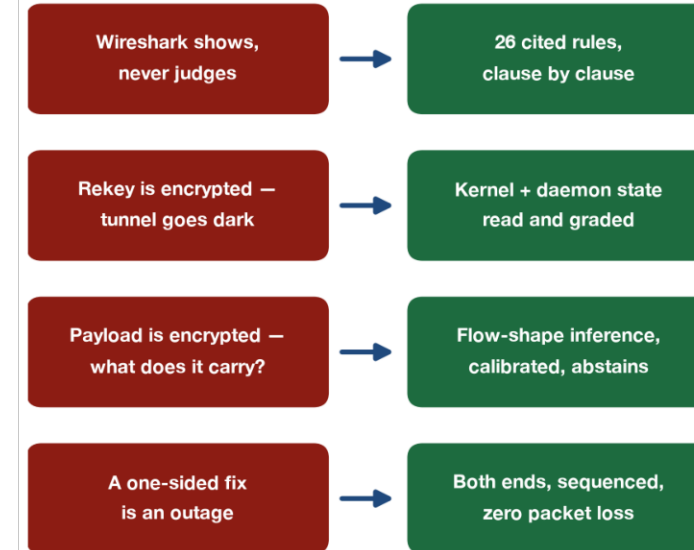
Innovation and uniqueness:

Facts are parsed. Estimates are inferred. A validator refuses to build a report that mixes them — the split is enforced, not promised.



GAP

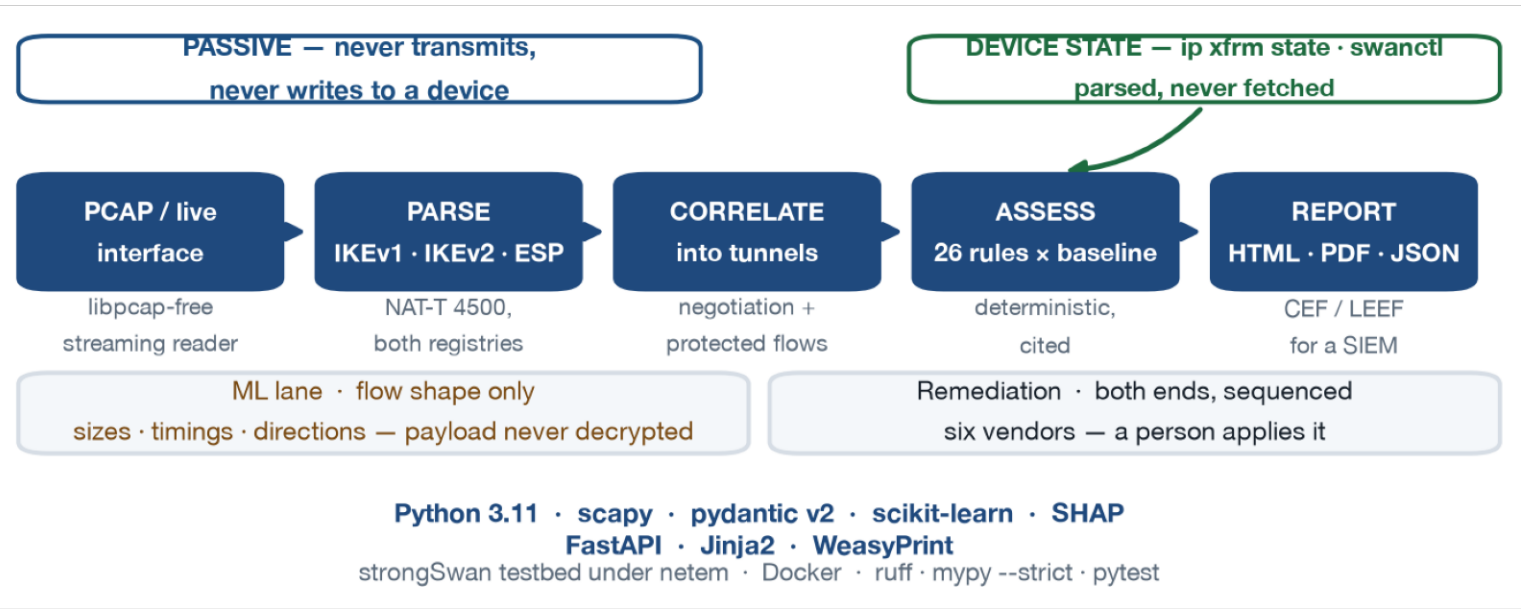
WHAT WE BUILT



Working prototype — v1.0.0 tagged, M11 acceptance 20/20, nothing skipped

Dashboard, demo video and source: github.com/PrashamJ17/SIH_PS160

◆ Technologies and methodology



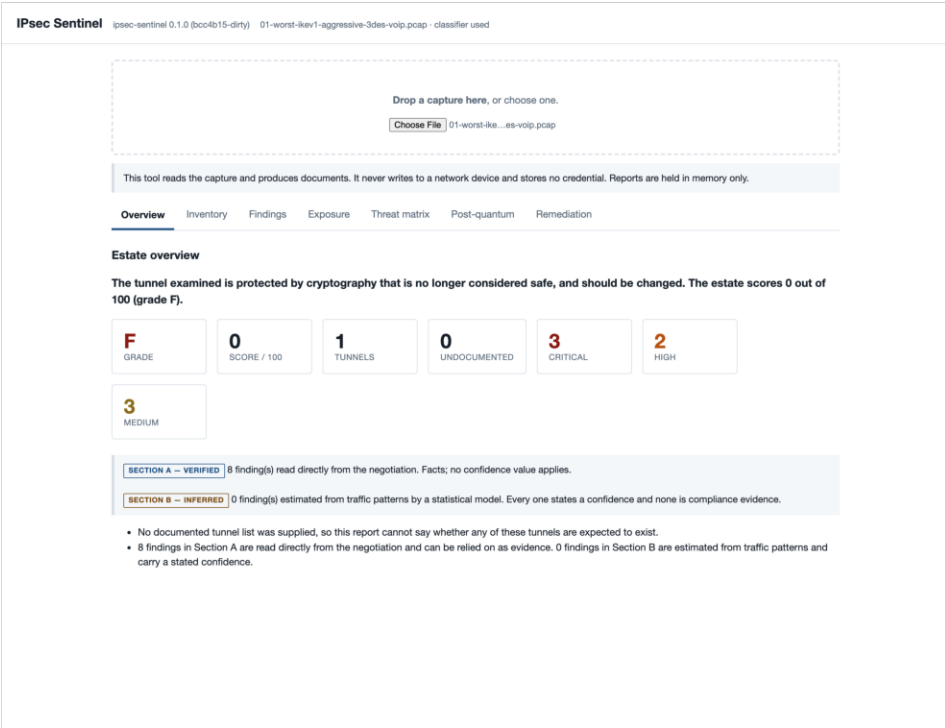
26 rules

6 baselines + 2 tags

6 vendor generators

IKEv1 · IKEv2 · ESP

Two boundaries the design will not cross:
It never writes to a network device and stores no credential — an AST sweep over every module enforces both. Configuration is generated; a person applies it.



Working prototype — the dashboard on a real capture

Reproducible build — git 5da6698

◆ Feasibility, the risks, and the strategies for overcoming them

FEASIBILITY

- Working prototype, v1.0.0 tagged
- Runs on a laptop: 0.68 GB container
- No agent, no credential, no outage

VIABILITY

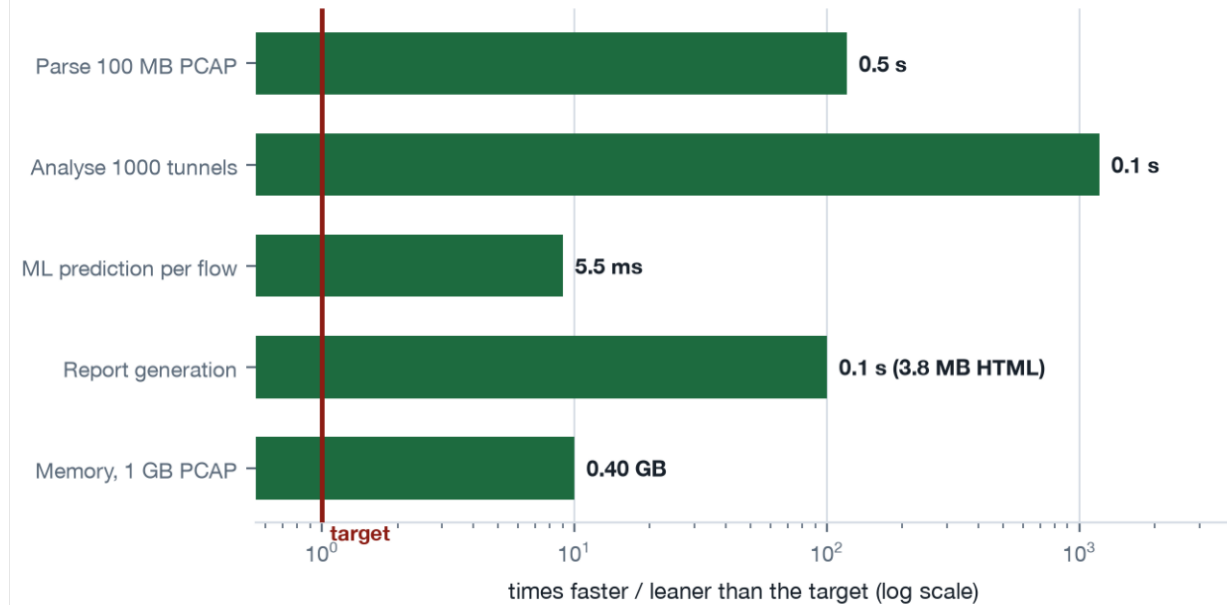
- Reads captures operators already have
- Air-gapped install, offline bundle
- ITSAR + CERT-In: no rival ships these

CHALLENGES & RISKS

- Model trained on laboratory traffic
- 5 of 6 vendor configs not device-tested
- A capture can miss an encrypted rekey

STRATEGIES

- Publish held-out numbers, worst included
- Every generated file states its own status
- Read kernel + daemon state to close it



2,613

unit tests

306

integration, on live pairs

93.8%

coverage

20/20

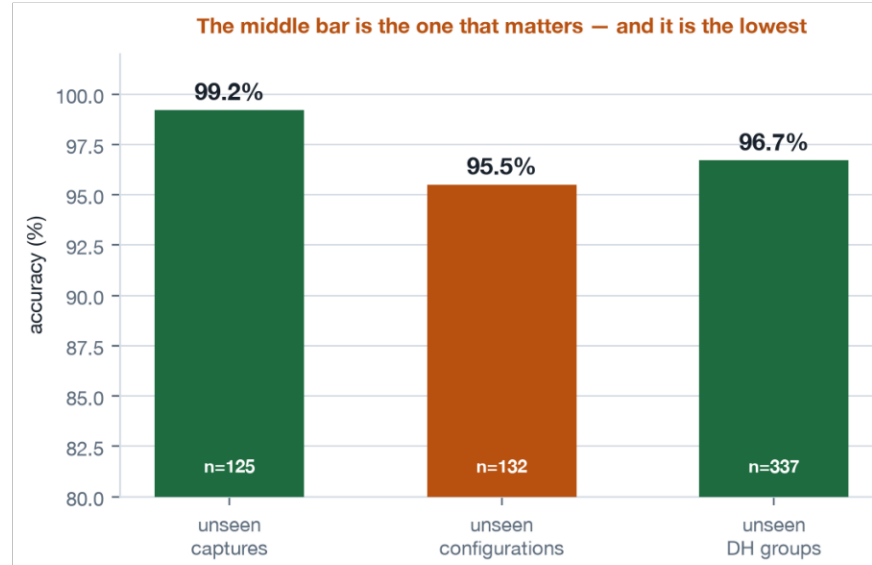
M11 acceptance

Deployable today:

0.68 GB container, non-root, runs under --network none.

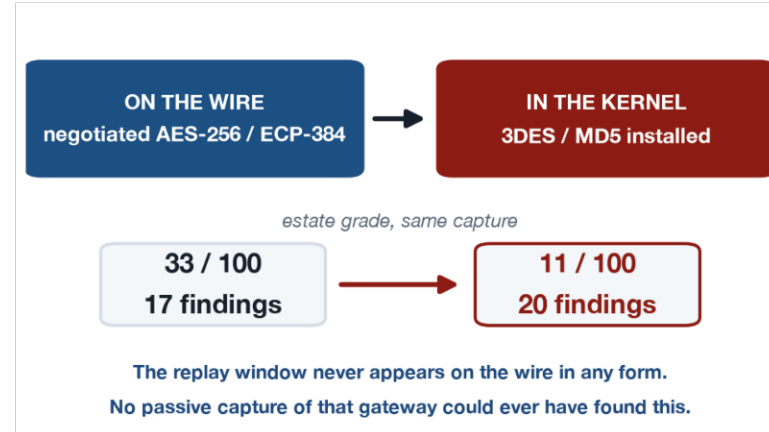
◆ Impact on the target audience, and the benefits

- 1 **Auditors**
cited evidence, not an opaque score
- 2 **Network teams**
both-ends fix, zero packet loss
- 3 **Regulators**
ITSAR & CERT-In from one capture
- 4 **Air-gapped sites**
runs offline, no credentials



Honest about its own limits

Trained on 637 rows across 7 classes of laboratory traffic. The unflattering number is published, not the best one.



Endpoint visibility

A gateway's own kernel state is parsed and graded against the same rules — never fetched, never with a credential.

Social and economic benefit:

Government, telecom and enterprise VPN estates get audit-grade evidence from a capture they already have — no agent, no credential, no outage, and an Indian baseline (ITSAR, CERT-In) that no comparable tool ships.



RESEARCH AND REFERENCES



◆ Details / Links of the reference and research work

RFC 7296

IKEv2 — the protocol the parser implements

RFC 8221 / 8247

Algorithm implementation requirements for ESP and IKEv2

RFC 9370 / 8784

Post-quantum: additional key exchanges, PPKs

NIST SP 800-131A Rev. 2

3DES disallowed after 2023; DES, MD5, SHA-1

ITSAR (NCCS) · CERT-In

Indian telecom security assurance requirements

RFC 2408 / 2409

ISAKMP and IKEv1, including aggressive mode

RFC 4303

ESP, and the anti-replay window SA-03/SA-04 judge

NIST SP 800-77 Rev. 1

IPsec VPN guidance — the default baseline

CNSA 1.0 · BSI TR-02102-3

The strict and German baselines

MITRE ATT&CK

Technique mapping for every finding

Our own published research:

GENERALISATION.md — held-out results including the lowest. CONFOUND_AUDIT.md — did the model learn traffic or cipher? DATASET.md — methodology and limits. LIMITATIONS.md — what it cannot do. external_dataset_audit.md — the named datasets contain no IPsec.

Code, evidence and demo:

github.com/PrashamJ17/SIH_PS160 — v1.0.0, M11 acceptance 20/20. Testbed: real strongSwan pairs under netem. Published JSON report schema 1.3. docs/SECURITY.md names the test behind every security claim.